

MOHAMMAD RAHMANITALAB

B.Sc. Mechanical Engineering (expected 2026), Sharif University of Technology

Minor Computer Science (expected 2026), Sharif University of Technology

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EDUCATION

B.Sc. in Mechanical Engineering, Sharif University of Technology Tehran, Iran
2021 – 2026 (expected)

- Advisor: Prof. Saeed Behzadipour
- GPA: 3.95/4.00 (18.93/20)
- Eligible for direct M.Sc. admission at Sharif University of Technology based on academic excellence.

Minor in Computer Science, Sharif University of Technology Tehran, Iran

Diploma in Mathematics and Physics, Payam Rastegaran High School Tehran, Iran
2021

- GPA: 4.00/4.00 (19.88/20)

INTERESTS

- **Programming:** Back End and Front End development, Microcontroller Programming
- **Machine Learning:** Neural Networks, Reinforcement Learning, Deep Learning
- **Robotics:** Exoskeletons, Surgical Robots, Humanoid Robots, Manipulators

SKILLS

Languages: Persian (Native), English (Excellent).

Programming: Java (advanced), C (advanced), Go (intermediate), C++ (intermediate), Python (intermediate), TwinCAT 3 (intermediate)

Simulation: MATLAB (advanced), ROS (beginner), Gazebo (beginner)

CAD and Dynamics: SolidWorks, MSC ADAMS

Soft Skills: Clean coding, teamwork, public speaking

UNDERGRADUATE RESEARCH PROJECT

Motion Analysis | Sharif University of Technology Present

- Performing motion analysis using IMU sensors to develop therapeutic protocols for children with Duchenne Muscular Dystrophy (DMD).
- Leading data collection, preprocessing, feature extraction, and applying machine learning and deep learning models for movement classification (MATLAB, Python).

TEACHING ASSISTANT EXPERIENCE

Automatic Control | Sharif University of Technology Spring 2025

- Supported instruction in control theory, including stability analysis, feedback systems.
- Contributed to grading assignments, quizzes, and exams while providing academic guidance to students.

Fundamental Automatic Control Design | Sharif University of Technology Fall 2025

- Supported instruction in controller design, including stability analysis, and PID design methods.
- Contributed to grading assignments, quizzes, and exams while providing academic guidance to students.

PROFESSIONAL EXPERIENCE

Back-End Developer | Moj Gostaran Adak, Tehran, Iran Oct 2025 – Dec 2025

- Developed a Java-based backend server integrated with the Traccar GPS tracking platform, enabling reliable device communication and data handling.
- Designed and implemented a customized Go backend for the company's internal frontend, supporting a locker-tracking system used in the Hamrah Aval project.
- Improved system functionality by building optimized APIs and ensuring seamless interoperability between tracking services and the client application.

INTERNSHIPS

Robotics | Sina Robotics and Medical Innovators Co, Tehran, Iran Jun 2024 – Sep 2024

- Worked on surgical robotics with emphasis on kinematics; developed analytical solutions for master-slave robotic systems to improve precision and responsiveness (MATLAB, MSC ADAMS, TwinCAT 3).
- Contributed to the design and testing of robotic components, focusing on responsiveness, stability, and haptic feedback in surgical robotic systems.

AWARDS AND HONORS

- Ranked 187th in the National University Entrance Exam (Konkur). 2021
- Ranked 85th in the National Master's Entrance Exam (Konkur), Mechanical Engineering. 2025
- Ranked 11th (top 11%) among 116 students in the Mechanical Engineering Department. Present
- Qualified for direct admission to M.Sc. programs ("without exam") due to academic excellence. 2025 & 2026

COURSE PROJECTS

Programming

Advanced Programming (Java)

Spring 2025

Developed an online multiplayer game using object-oriented design, multithreading, and network programming.

Fundamentals of Programming (C)

Spring 2022

Implemented a Twitter-like social media application including user management and data persistence.

Robotics and Control

Robotics

Fall 2024

Modeled and simulated a robotic manipulator in MATLAB/Simulink to study kinematics and dynamic behavior.

Fundamental Automatic Control Design

Fall 2024

Designed and tuned a PID controller for a ball-and-beam system using MATLAB/Simulink.

Automatic Control

Spring 2024

Designed a PID controller for a rotating governor and analyzed system performance and stability.

Dynamics of Machines

Spring 2024

Simulated a pushing mechanism in MATLAB/Simulink to analyze forces, motion, and dynamic response.

SELECTED COURSES

Completed:

- Fundamentals of Programming (C)
- Advanced Programming (Java)
- Discrete Mathematics
- Probability
- Dynamics of Machines
- Robotics
- Automatic Control
- Fundamental Automatic Control Design

Ongoing:

- Linear Algebra
- Operating Systems
- Data Transmission and Networks
- Computer Simulation

OTHER EXPERIENCE

Website Developer: Developed and maintained the website for ICERS, Mechanical Engineering Department (WordPress).

EXTRA-CURRICULAR ACTIVITIES

ICERS 2024: Contributed to event organization and media coordination as a member of the IT and Media Team.

CARE Student Society: Served as Media Director, managing media content and team operations.

Mechanical Engineering Department: Website developer for departmental web presence.